Tender: Development of Centralized Access Control Integration System — B. BIRDS (Border Incident Reporting & Documentation System) - Phase II
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PERFORMANCE TEST PLAN

BIRDS (Border Incident Reporting & Documentation System) - Phase II

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1. Purpose and scope

This plan describes how the supplier proves, on BGB's installed infrastructure and before user acceptance testing, that BIRDS meets the performance and capacity requirements of the Technical Specification:

Item	Requirement	How this plan tests it
37	Scalability	Capacity run beyond 500 users (run P9) and a comparison of one and two application servers
38	Concurrent users [100/250/500]	Load runs at 100, 250 and 500 concurrent users (P2–P4); all core functions checked during the run
39	Response within 3 seconds	Screen and request times measured at the users' side in every run; acceptance criteria in section 5
40	Availability 99.5 %	Endurance run over one shift (P7) and failover under load (P8); availability itself is measured in operation

Item 38 leaves the figure open ([100/250/500]). The plan tests all three; acceptance is judged at the figure BGB confirms, and at 500 if no figure is confirmed. The results are given to BGB in a signed test report (section 8). Large file uploads and downloads and complex report generation are measured and reported, but are excluded from the 3-second criterion, as item 39 states.

2. Test environment

2.1 Servers and configuration

The test runs on the infrastructure installed for production (Technical Proposal, section 7.2), configured exactly as for go-live, so that the result applies to the system BGB will use:

a. Load balancers with HTTPS, two application servers with all processes running, PostgreSQL primary with the standby replica active, central file storage, and the antivirus engine checking uploads.

b. Production settings: session timeout, password policy, rate limits, logging and audit exactly as they will be on go-live; nothing is switched off to make the test faster.

c. A separate test database on the production servers (not BGB's live data), removed after the test together with the test accounts.

2.2 Data volume

Response times depend on how much data the database holds, so the test database is filled with three years of realistic data before the runs, following the workload basis of the Technical Proposal (section 7.1): about 60,000 incidents across all units with their history, comments and evidence records, a sample of real evidence files, and an audit trail, messages and notifications of the matching size. If BGB migrates existing records (item 56), the migrated volume is used if it is larger.

2.3 Load generators and monitoring

a. Two computers on BGB's network, in the place where users connect from, run the simulated users; they are not the servers under test.

b. Monitored on every server during each run: processor, memory, disk and network use; PostgreSQL connections, locks and slow statements; application and error logs.

c. Times are measured at the users' side, so they include the network, the load balancer and HTTPS, as users experience them.

2.4 Before the test starts

a. Functional testing completed; the software version and configuration under test are recorded and frozen.

b. Backups, replication and monitoring working; BGB IT informed, as the test produces load on the network.

c. Test accounts created for every role in the test database (section 4.1); a short trial with 10 users confirms that the tool, the accounts and the monitoring work.

3. Test tool

The supplier's load tool is part of the delivered source code (server/loadtest), so BGB can repeat the test at any time, for example after an upgrade. It is already built and was used for the dry run in Annex A.

a. Each simulated user signs in with its own account (BIRDS allows one session per user) and keeps the realtime connection open, as the web application does.

b. Screens send exactly the requests the web application sends, recorded from the browser, in parallel as a browser does; evidence previews are downloaded, decrypted and integrity-checked like on screen.

c. Background checks run on the application's own timers; sessions are renewed when the 15-minute access token expires.

d. It records every request and every screen with its time and result, and the server's processor and memory use; results are saved for the report.

4. Workload model

4.1 Users

The simulated users reflect how BIRDS is used: most users file reports at battalions and BOPs, fewer review them in the sectors, and a small number work at HQ.

Role	Share	Scope
Battalion Operator	60 %	Spread over all battalions
Sector Officer	15 %	Spread over all sectors
Verifier/Approver	10 %	Spread over all sectors
Auditor	10 %	HQ
HQ Administrator	5 %	HQ

4.2 What each user does

Each user opens a screen, reads or works on it, and opens the next. The time between screens varies at random around 20 seconds on average (normal pace) or 7 seconds (peak pace). Users open only the screens their role permits, in these proportions:

Screen	Share	Requests to the server
Dashboard	25 %	11: key figures, two trends, two breakdowns, recent incidents, five work-queue counts
Incident register	20 %	3: a page of 25 incidents, sometimes filtered or searched, and two counts
Incident record	25 %	2 to 4: the record with its timeline and evidence list, related incidents, up to two evidence previews
GIS map	8 %	1: incident locations in the user's scope
Analytics	7 %	8: key figures, trends for this and last year, five breakdowns
Reports	5 %	1: the operational report
Messages	5 %	1: the user's conversations
Flash alerts	3 %	1: the active alerts
Add a comment	2 %	1: a comment on an incident (writes to the database and the audit trail)

In addition every user keeps one realtime connection open and makes the background checks of the application: active alerts and notifications every 30 seconds, unread messages every 60 seconds. For the test at BGB the supplier adds the other daily work to the mix in its normal proportions: filing new incidents with evidence files, workflow actions (review, verify, approve, return), exports and printouts.

5. Test runs

Run	Name	Users	Duration	Purpose
P1	Baseline	10	15 min	Reference times with almost no load; checks the set-up
P2	Load 100	100	30 min	Item 38, first figure
P3	Load 250	250	30 min	Item 38, second figure
P4	Load 500	500	60 min	Item 38, highest figure; the main acceptance run for item 39
P5	Peak	500, three times as active	30 min	Shift change or a major incident, when most users are busy at once
P6	Sign-in surge	500 signing in within 5 min	15 min	Start of a shift: the time to sign in and open the dashboard
P7	Endurance	500	8 hours	One full shift: response times stay stable, no growth in memory or errors (item 40)

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Run	Name	Users	Duration	Purpose
P8	Failover under load	500	60 min	One application server stopped, then the database standby promoted: service continues (items 36, 40)
P9	Capacity	600, 700 … in steps of 100	Until a limit	How far beyond 500 users the installation goes (item 37); for information

Each run starts with users signing in over a few minutes; the measurement period begins when all have signed in. During every run a tester also works in BIRDS by hand (files an incident, approves one, sends a message, prints a report) to confirm that no core function degrades (item 38).

6. Acceptance criteria

Measure	Criterion
Response time (item 39)	In runs P2, P3 and P4, 95 % of normal screens complete within 3 seconds, and the median within 1 second. Normal screens: sign-in, dashboard, incident register, incident record, map, analytics, messages, flash alerts, saving a form, workflow actions.
Errors	Fewer than 0.1 % of requests fail (server errors or no answer); no simulated user fails to sign in.
Core functions (item 38)	All checks by hand during P4 succeed; realtime messages arrive within 2 seconds.
Reserve	In P4, average processor use below 70 % on the application servers and on the database server.
Endurance (P7)	Response times in the last hour within 20 % of the first hour; memory use of the application stable; no errors building up.
Failover (P8)	Stopping one application server: the service continues and fewer than 1 % of requests fail during the switch. Promoting the database standby: service restored within the recovery time of the Technical Proposal (section 8).
Excluded from the 3 s criterion	Uploading and downloading large files, Excel and CSV exports, printouts and other complex reports: measured and reported with their times.

7. Procedure and schedule

The performance test is part of the testing phase of the work plan (Technical Proposal, section 11, weeks 6–8), after the installation and functional testing and before user acceptance testing.

Day	Work
Day 1	Load the test data, create test accounts, check monitoring; run P1 and P2

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Day	Work
Day 2	Runs P3, P4 and P5; first analysis
Day 3	Runs P6 and P8; P7 overnight
Day 4	Run P9; analysis; draft report
Reserve	Two days for tuning and repeat runs if a criterion is missed (section 9)

Who	Responsibility
Supplier: performance and database engineers	Prepare the data and the tool, run the tests, monitor, analyse, tune, write the report
BGB IT	Provide access, observe the runs, confirm the network conditions, sign the report
BGB nominated officers (optional)	Do the checks by hand during the runs

8. Reporting

The performance test report is given to BGB within three working days of the last run and signed by both parties. It contains:

a. The environment as tested: servers, versions, configuration and data volume.

b. For every run: users, duration, requests per second, response times (median, 95 % and 99 %) per screen and per request type, errors, and processor, memory and disk use of each server, with graphs over time.

c. The checks by hand during the runs and their results.

d. The result against each criterion of section 5, findings, tuning carried out, and the capacity found in P9.

e. The raw results from the tool, so that BGB can check or repeat the figures.

9. If a criterion is not met

a. The supplier finds the cause from the measurements: slow database statements, processor or memory limits, connection limits, network.

b. It corrects it at no cost to BGB: query or index changes, application changes, configuration (processes, connection pools, caches).

c. The run is repeated and the report records the change and both results.

d. If the cause lies in infrastructure that differs from the specification in the Technical Proposal (section 7), the supplier documents it and agrees the remedy with BGB.

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Annex A. Dry run on the demonstration build

The dry-run results are not yet available.

Name of authorised signatory:	
Designation:	
Signature and seal:	
Date:	18/10/2026

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